

AI Engineering Summer Training Daily Activity Log

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Week: 1

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Instructions

Complete this document at the end of every training day. At the end of each week, upload the updated document to the shared Google Drive folder.

When you upload the file to drive rename from `AI_Engineering_Daily_Project_Log_Template.docx` to `FirstName_ID_WeekNum_docx`

Day 1 Date: Jul/05/2026 Today's Topic: Course intro & how LLMs work

Daily Task Completed:

- VsCode, python, git, and LLM API setup completed, and first API request was successful
- Tested 2 different prompts
- Tested different API parameters such as temperature and max output tokens.

What I Learned (3-5 points):

- I learned how LLMs works and discovered that while cloud APIs like Gemini are fast, hosting locally with Ollama is a great free alternative for unlimited testing despite being hardware dependent.
- I learned the purpose of git add, git commit, and git push.
- I understood that the API does not remember previous conversations unless I provide the conversation history.
- I also learned how Python virtual environments keep project dependencies isolated and safe from system conflicts.
- I practiced adjusting the temperature setting to see how it directly changes the model's creativity and randomness with the same prompt.

Work Evidence (GitHub / Colab / Files):

<https://github.com/MkJamil/O-LLM>

Challenges Faced:

- I faced some challenges with the LLM first i tried to use Google Gemini API for my task but I ran out of tokens quickly from just testing random prompts with different temperatures/parameters

How I Solved Them:

- I switched to Ollama for unlimited access but i has some problems with performance after some basic research I tried changing to smaller model from llama3 which has 3 billion parameters so I switch to qwen 2.5 which has 0.5B parameters

Self-Evaluation (Excellent / Good / Fair / Need More Practice):

Good

Supervisor Comments:

Day 2 Date: Jul/06/2026 Today's Topic: Prompting Fundamentals

Daily Task Completed:

- Learned what prompt engineering is and how to write better prompts
 - Tested 5 different prompts with the same task but with different parameters and constraints
 - Used different prompt components such as Role, Task, Constraints, and Format.
 - Changed the way I interact with the LLM from just printing answers to chatbot style
 - Switched from locally hosing Ollama to using docker
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What I Learned (3-5 points):

- Learned the four main components of a good prompt: Role, Task, Constraints, and Output Format.
- I learned that prompt structure heavily impacts the quality and variety of an LLM's output when generating ideas.
- By testing five different prompt styles ranging from a simple request to highly detailed instructions, I saw how each one produced very distinct results.
- I learned how containerization with Docker keeps the development project completely clean and isolated.
- I realized that giving the model a clear persona (like a software engineer or a CEO) helps tailor the perspective of the answer.

Work Evidence (GitHub / Colab / Files):

[https://github.com/Mk\]amil/O-LLM](https://github.com/Mk]amil/O-LLM)



Prompt
engineering-Task2.docx

Challenges Faced:

- The biggest challenge was switching from local Ollama to docker and figuring out the setup
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How I Solved Them:

- I carefully went through the terminal logs to debug connection issues between my local environment and the Docker container.
 - I adjusted the container settings to make sure it was utilizing my CPU hardware efficiently for model execution.
 - I practiced using Docker commands to start, stop, and verify that the Ollama service was running smoothly inside the background container.
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Self-Evaluation:

Fair

Supervisor Comments:

Day 3 Date: Jul/07/2026 Today's Topic: AI Development Toolchain

Daily Task Completed:

- Explored the core tools of the AI development toolchain, including APIs, CLIs, and IDEs.
- Defined how an API acts as the communication bridge between custom code and an AI model or server.
- Learned how a CLI uses direct terminal commands to interact with the computer faster and more powerfully than a standard GUI.
- Analyzed and mapped out the full AI development workflow diagram showing how a prompt moves from the user, through the application and API, to the LLM and back.

What I Learned (3–5 points):

- I learned how a prompt actually travels from the user through the application and the API to get a response from the LLM.
- Visualizing the workflow diagram really helped me finally understand the API, which was the concept I was having the most trouble with.
- Seeing the entire chain helped solidify everything I learned from Day 2 about how prompts interact with the application.
- I discovered that the CLI feels much more powerful and fast even though it is less beginner-friendly, whereas a GUI is easier to use but feels restricted.

Work Evidence (GitHub / Colab / Files):



AI Development
Toolchain.docx

Challenges Faced:

- I was having trouble understanding the API workflow and how it actually works

How I Solved Them:

- The diagram helped me understand what is the biggest difference between GUI is, CLI and API

Self-Evaluation (Excellent / Good / Fair / Need More Practice):

Excellent

Supervisor Comments:

Day 4 Date: Jul/08/2026 Today's Topic: AI Workflow

Daily Task Completed:

- Initiated the development of a complete AI workflow deployed inside a Docker container.
- Configured the workflow to accept text inputs and process them automatically for text summarization.
- Integrated conditional logic inside the pipeline to dynamically choose the output format based on the prompt's instructions.
- Developed the capabilities to generate the final summary as either a standard text file (summarize.txt) or a formatted A4-sized PDF file depending on what the user explicitly requests.

What I Learned (3–5 points):

- I learned how to implement prompt-driven output, which was tricky but really interesting since it forces the LLM to dynamically choose the file format based on user instructions.
- I discovered that because "literally everything" runs inside the Docker container, any generated files (like the PDF or text file) are created inside the container's isolated environment.
- I learned how to set up an output directory and manage file paths so that the generated files can be successfully copied or accessed outside of the container.

Work Evidence (GitHub / Colab / Files):

[https://github.com/Mk\[amil/O-LLM](https://github.com/Mk[amil/O-LLM)

Challenges Faced:

- I faced challenges in the prompt driven generation it was confusing at first to make the AI understand that the generation command was not part of the text

How I Solved Them:

- The solution was to create a certain phrase and make the python script read through the prompt if the certain phrase was found then it should initiate the generation process and to generate depending on that phrase to either pdf or txt

Self-Evaluation (Excellent / Good / Fair / Need More Practice):

Need more practice

Supervisor Comments:

Day 5 Date: Jul/09/2026 Today's Topic: AI Workflow + Documentation

Daily Task Completed:

- Refined and optimized the prompt-driven AI summarization workflow inside the Docker container to ensure consistent output formatting.
- Completed and structured the formal technical documentation for the entire week's tasks and learning modules.

What I Learned (3–5 points):

- I realized how valuable it is to step back and document my work, as reviewing my daily tasks step-by-step helped solidify my understanding of the entire tool chain and local environment setup.

Work Evidence (GitHub / Colab / Files):

<https://github.com/MkIamil/O-LLM>

Challenges Faced:

- This past week was an entirely new experience that presented several unique and interesting technical challenges.

How I Solved Them:

- I carefully refined my prompt structure and conditional logic until the LLM could consistently and accurately choose the correct file format.
- I resolved the isolated container file issue by setting up a dedicated output directory to properly copy and access the generated PDFs and text files outside of Docker.

Self-Evaluation (Excellent / Good / Fair / Need More Practice):

Good

Supervisor Comments:

Weekly Reflection

What was the most valuable thing you learned this week?

The most valuable takeaway was understanding the end-to-end AI toolchain and realizing how containerization with Docker keeps projects clean and portable. Seeing the workflow diagram completely clarified how an API bridges custom code to the LLM and learning that a Docker container allows anyone else to run my model instantly without dealing with environment setup issues was a massive eye-opener.

What was the biggest challenge?

The biggest challenge was moving away from running Ollama natively to setting up and configuring everything inside a Docker container

Which topic do you still need to practice?

Docker in general the ideas behind the containers and images while the setup is still taunting, I still need to practice working with the CLI and terminal commands since it feels incredibly fast and powerful but is less beginner-friendly and harder to navigate than a GUI.

Additional Notes:

Building this workflow proved how useful it is to host models locally and inside containers. It means the entire environment is self-contained, making the AI application reliable, modular, and easy to share with anyone else without breaking things.

Weekly Submission Checklist

- ✓ Daily log completed
- ✓ Code uploaded
- ✓ GitHub updated
- ✓ Required files included
- ✓ Weekly reflection completed
- ✓ Uploaded to Google Drive